

Exact Excitation Blocks, Propagators, and Revivals in the Jaynes–Cummings Model

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Abstract

We give a convention-complete spectral and dynamical atlas for the single-mode rotating-wave Jaynes–Cummings Hamiltonian. Total excitation decomposes the atom–Fock space into a separate vacuum and exact two-dimensional blocks. We derive every dressed pair, the full block propagator, transition probability, trace and determinant. For a finite active set, universal bare-population revival is equivalent to rational commensurability of its positive Rabi frequencies; full state-vector revival requires additional center and vacuum phase alignment. Resonance, zero coupling, block indexing, coupling-sign gauge, finite cutoffs and the noncompact full-Fock unitary are kept explicit. Independent exact/high-precision, symbolic, replay and hostile-mutation audits accompany the theorem.

1 Hamiltonian and excitation convention

Set $\hbar = 1$ and

$$H = \omega_c a^\dagger a + \frac{\omega_q}{2} \sigma_z + g(a^\dagger \sigma_- + a \sigma_+), \quad N = a^\dagger a + \sigma_+ \sigma_- . \quad (1)$$

We use $\sigma_z|e\rangle = |e\rangle$ and $\sigma_z|g\rangle = -|g\rangle$. For $n \geq 1$, excitation block n has the ordered basis $\{|e, n-1\rangle, |g, n\rangle\}$; this shift is part of the frozen convention.

Theorem 1 (Conserved-excitation atlas). *The operators in (1) satisfy $[H, N] = 0$. The vacuum $|g, 0\rangle$ is an eigenstate with energy $-\omega_q/2$. On block $n \geq 1$,*

$$H_n = c_n I + \frac{\Delta}{2} \sigma_z + g\sqrt{n} \sigma_x, \quad c_n = (n - \tfrac{1}{2})\omega_c, \quad \Delta = \omega_q - \omega_c. \quad (2)$$

Let $\Omega_n = (\Delta^2 + 4g^2n)^{1/2}$. The dressed energies are

$$E_{n,\pm} = c_n \pm \frac{\Omega_n}{2}, \quad (3)$$

and the exact propagator is

$$U_n(t) = e^{-ic_nt} \left[\cos \frac{\Omega_n t}{2} I - i \frac{\sin(\Omega_n t/2)}{\Omega_n} (\Delta \sigma_z + 2g\sqrt{n} \sigma_x) \right], \quad (4)$$

with its continuous identity limit when $\Omega_n = 0$. In particular,

$$P_{|e,n-1\rangle \rightarrow |g,n\rangle}(t) = \frac{4g^2n}{\Omega_n^2} \sin^2 \frac{\Omega_n t}{2}, \quad (5)$$

where the simultaneous $g = \Delta = 0$ value is zero. Every U_n is unitary, and

$$\text{tr } U_n = 2e^{-ic_nt} \cos(\Omega_n t/2), \quad \det U_n = e^{-2ic_nt}. \quad (6)$$

Proof. Each interaction monomial creates one photon while destroying one atomic excitation, or reverses that operation, proving $[H, N] = 0$. Applying H to the ordered bare basis yields (2). If $B_n = H_n - c_n I$, then $B_n^2 = (\Omega_n^2/4)I$. The characteristic polynomial gives (3), while the even and odd parts of e^{-itB_n} give (4), including the continuous zero-frequency limit. Its off-diagonal modulus gives (5). The Pauli-square identity gives unitarity; its two eigenphases give (6). $\square$

2 Finite-support revival theorem

Call a block active and coupled when the initial support includes it and $g\sqrt{n} \neq 0$.

Proposition 1 (Population versus state-vector revival). *For a finite set S of active coupled blocks, there is a time $T > 0$ that restores every bare population for every bare initial state in those blocks if and only if*

$$\Omega_n T \in 2\pi\mathbb{Z} \quad (n \in S). \quad (7)$$

Equivalently, the positive frequencies $\{\Omega_n : n \in S\}$ are rationally commensurate. If $\Omega_n T = 2\pi k_n$, then $U_n(T) = e^{-ic_n T} (-1)^{k_n} I$. A common state-vector revival therefore also requires these phases, together with any occupied vacuum phase $e^{i\omega_q T/2}$, to agree up to one global phase.

Proof. For a coupled block the coefficient in (5) is positive, so universal bare-population restoration is equivalent to $\sin(\Omega_n T/2) = 0$. This is (7). A finite positive frequency set admits a common such T exactly when all frequency ratios are rational. Substitution in (4) gives the parity and center phase, proving the stronger-state clause. $\square$

3 Singular and operator boundaries

When $g = 0$, the bare states decouple; the case $g = \Delta = 0$ uses the continuous limit in (4), not a $0/0$ expression. At resonance, $\Omega_n = 2|g|\sqrt{n}$ and the exchange amplitude in (5) is one. The vacuum is one-dimensional and is never inserted into an $n \geq 1$ square-root formula. Conjugation by a diagonal atomic phase maps g to $-g$, so energies and probabilities agree although the signed off-diagonal amplitude changes.

A finite Fock compression retains every excitation block below its dangling top state. On the full infinite-dimensional space, a unitary maps an orthonormal sequence to another orthonormal sequence; it is therefore not compact, and its infinitely many singular values equal one. Thus it is not Schatten or trace class, and no ordinary full-space trace or Fredholm determinant is asserted.

Source note

Jaynes and Cummings [1] introduced the rotating-wave atom–mode owner. Shore and Knight [2] review its dressed dynamics and collapse/revival context. We claim no priority for those classical results; coherent-field asymptotic collapse and revival are outside the theorem.

References

- [1] E. T. Jaynes and F. W. Cummings, *Comparison of quantum and semiclassical radiation theories with application to the beam maser*, *Proceedings of the IEEE* 51 (1963), 89–109. DOI: 10.1109/PROC.1963.1664.
- [2] B. W. Shore and P. L. Knight, *The Jaynes–Cummings model*, *Journal of Modern Optics* 40 (1993), 1195–1238. DOI: 10.1080/09500349314551321.

Declarations

Scope. NO_BAD_EULER_OR_ROOT_NUMBER; no target arithmetic, target-zero or Hilbert–Polya claim.

Data and code. The theorem, ledger and independent audits are released with HCS-C223. **AI-use**

disclosure. Generative tools assisted drafting and code generation; the internal artifact chain checked the displayed claims and metadata. This is not external peer review.